



PharmOS

Community pharmacy, simplified.

ASSIGNMENT 1

Product Vision and Rapid Prototyping

Project scope

PharmOS is a desktop-first pharmacy management system for small community pharmacies. The project intentionally prioritizes a small number of reliable, well-executed workflows over a broad set of partially implemented features.

Project Team

Team member	Project role	Area of expertise
Hassan Khalil	Project Manager	Full-stack, ML, & embedded sys
Dana Shmait	Team Member	Front-end development
Nicolas Azar	Team Member	Embedded systems and hardware

Part One - Product Vision

1. Review of Existing Products

The review focuses on four established pharmacy management products whose official feature sets overlap with the daily needs of community and independent pharmacies. The goal is not to copy them, but to identify proven patterns and decide which ones are worth implementing in a smaller, more focused product.

Product	Main relevant features	What it demonstrates
PioneerRx [1]	Customizable workflow; inventory and ordering; integrated POS; reporting and financial tools; patient-facing and mobile tools.	A pharmacy platform can centralize many workflows, but its scope can become very broad.
Liberty PharmacyOne [2]	Flexible workflow and queues; inventory control and ordering; integrated checkout/POS; reporting and data views; patient communication.	Workflow speed, integrated POS, and visibility are core recurring needs.
PrimeRx [3]	Patient profiles and medication history; inventory management; POS; refill dashboard; reporting; workflow automation.	Dashboards and automation can reduce repetitive work when kept understandable.
BestRx [4]	Prescription processing; patient profiles; workflow automation; inventory management; POS; analytics and reporting.	Simple, centralized screens and automation are strong usability patterns for independent pharmacies.

2. Business Opportunity

The reviewed systems show a clear pattern: inventory, sales, records, workflow automation, and reporting are valuable. They also include many advanced modules such as insurance adjudication, multi-store management, clinical programs, delivery systems, and large integration ecosystems. PharmOS can target the opposite design problem: a small community pharmacy that wants the essential daily workflows in one dependable desktop application without enterprise-level complexity.

Opportunity statement

Build a focused pharmacy operating system that is fast to learn, works naturally with common pharmacy hardware, makes stock problems visible early, and keeps the most frequent tasks within a few clicks. PharmOS should compete on clarity and reliability, not feature count.

3. Features for PharmOS

Inventory and expiry control: Track medicines, quantity, batch/expiry information, low-stock thresholds, and barcode-based item lookup. Surface low-stock and near-expiry items immediately.

Fast sales / POS: Create a sale through barcode scan or search, calculate the total, record the transaction, print a receipt, and deduct stock automatically.

Customer and prescription log: Maintain a lightweight customer profile with searchable prescription and purchase history. This is a record-management feature, not a clinical diagnosis system.

Smart Stock Assistant: Use recent sales history and current stock to suggest likely shortages and reorder quantities. Suggestions remain advisory and require staff confirmation.

Dashboard and reports: Show only the information that needs attention: low stock, near-expiry items, sales activity, stock totals, and basic historical reports.

Scope control

To protect quality, the first development stages deliberately exclude features that would significantly increase complexity without improving the core workflow: insurance claim adjudication, external e-prescribing networks, clinical interaction diagnosis, delivery routing, patient mobile apps, and multi-store chain management. Security, backups, user permissions, and an audit trail are treated as baseline system requirements rather than extra headline features.

4. Product Vision

Vision statement

For small community pharmacies that need a dependable way to run daily operations, PharmOS is a desktop pharmacy management system that combines stock control, sales, customer/prescription records, and simple stock intelligence in one fast interface. Unlike broad enterprise pharmacy suites, PharmOS prioritizes a small number of workflows, local hardware integration, clarity, and reliability.

Product principles

- Reliable before clever - A feature is only valuable if it works consistently. Critical workflows must fail safely and preserve data.
- Few clicks for common work - Sales, stock lookup, and alerts should be immediately accessible from the homepage.
- Attention, not clutter - The dashboard should emphasize exceptions - low stock, expiry risk, and actionable tasks - instead of showing every available metric.
- Human-controlled automation - AI and automation may recommend actions, but stock orders and record changes remain visible and confirmable by staff.
- Hardware-friendly desktop design - Barcode scanners and receipt printers should fit naturally into the workflow without requiring complicated setup.

MVP success criteria

- a) A staff member can start and complete a normal sale without leaving the POS workflow.
- b) Adding or scanning a medicine immediately updates or retrieves the correct inventory record.
- c) Low-stock and expiry warnings are visible from the homepage and link directly to the relevant items.
- d) Smart Stock suggestions are based on stored sales/inventory data and can never place an order automatically.
- e) The system can back up and restore its local data without losing transaction history.

Primary users

The primary users are pharmacists and pharmacy staff in a single small community pharmacy. The interface assumes frequent desktop use at the counter and behind the pharmacy desk, with keyboard, mouse, barcode scanner, and receipt printer access.

Part Two - Rapid Prototyping

The supplied Figma tutorial emphasizes using the prototype to visualize structure, navigation, buttons, and feasibility rather than treating it as a finished coded interface. It also recommends a desktop frame for software intended for computer use. [5] The PharmOS prototype follows that approach.

Homepage prototype

The homepage is a desktop dashboard. Its first job is to show what requires attention and provide direct access to the core modules. All numbers shown below are sample prototype data.

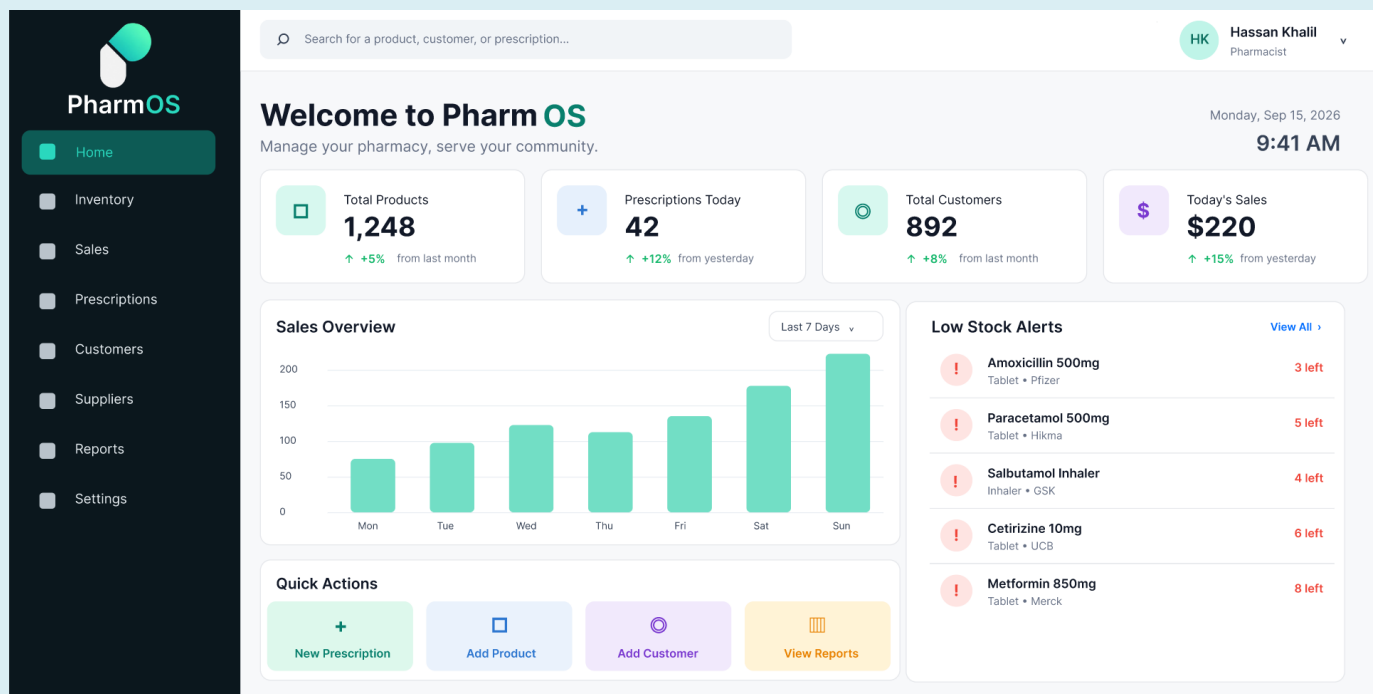


Figure 1. PharmOS desktop homepage prototype - sample data.

[Figma Diagram Access Link](#)

(very much worth checking out)

Prototype Interaction Plan

The prototype is intentionally simple. Each visible button or module has a clear destination so later assignments can turn the design into a working flow without redesigning the homepage.

Homepage element	Destination	Purpose
New Sale	Sales / POS	Starts the most common transaction workflow immediately.
Inventory / View inventory	Inventory	Search stock, add medicines, review quantity, batch, and expiry information.
Customers & Rx	Customer records	Open customer profiles and prescription/purchase history.
Smart Stock	Stock suggestions	Review predicted shortages and suggested reorder quantities before confirming any action.
Reports	Reports	View basic sales and inventory history.
Alert cards	Filtered inventory	Jump directly to low-stock or near-expiry items instead of searching manually.

Why this homepage structure?

- The left navigation keeps every core module visible without nested menus.
- The top area gives one obvious primary action - New Sale - instead of several competing buttons.
- Low-stock and expiry cards convert inventory data into immediate tasks.
- The Smart Stock panel is visible but secondary, reinforcing that AI assists the pharmacist rather than controlling the workflow.
- The design uses a standard widescreen desktop ratio and compact cards so it works naturally on common pharmacy PCs.

References

- [1] PioneerRx - Pharmacy Software - <https://www.pioneerrx.com/pharmacy-software>
- [2] Liberty Software - PharmacyOne - <https://libertysoftware.com/pharmacy/>
- [3] PrimeRx - Pharmacy Management Software - <https://www.primerx.io/primerx-software>
- [4] BestRx - Pharmacy Software - <https://www.bestrx.com/pharmacy-software>
- [5] Introduction to Figma: Building Your First Design System